Skill Gap Analysis and Personalized Roadmap Generator

A Mini project Work Submitted in Partial fulfilment of the requirement

For the award of the Degree of

BACHELOR OF SCIENCE (COMPUTER SCIENCE WITH DATA SCIENCE)

Submitted by

AMALA JENIFA A

REG.NO: 24081231802912103

PESILA V

REG NO: 24081231802912125

Under the Guidance of

Dr. S. Narasimma Subramanian M.C.A., M.Phil., Ph.D.

DEPARTMENT OF COMPUTER SCIENCE

SRI PARAMAKALYANI COLLEGE

(AFFILIATED TO MANONMANIAM SUNDARANAR UNIVERSITY TIRUNELVELI)

OCTOBER - 2026



SRI PARAMAKALYANI COLLEGE

ALWARKURICHI – 627 412

CERTIFICATE

The project work embodied in the present Dissertation entitled **"SKILL GAP ANALYSIS AND PERSONALIZED ROADAP GENERATOR"** is a bonafide work done by A. AMALA JENIFA (REG.NO: 24081231802912103), V. PESILA (REG.NO: 24081231802912125) Department of Computer Science, Sri Paramakalyani College, Alwarkurichi, Tenkasi – 627412 has been carried out in partial fulfilment of the award of the Degree of the BACHELOR OF SCIENCE IN COMPUTER SCIENCE WITH DATA SCIENCE in during the Academic Year 2025-2026.

HEAD OF THE DEPARTMENT **GUIDE**

Submitted for the University Examination held on: _____

INTERNAL EXAMINER EXTERNAL EXAMINER

ABSTRACT

The **Skill Gap Analysis & Personalized Roadmap Generator** is a web-based career guidance system developed to help students and job seekers understand their career requirements and identify the skills needed to achieve their desired career goals. Users can create and manage their profiles, provide their educational and skill information, and select their preferred career role. The system uses a career-skill dataset to identify the skills required for the selected career and compares them with the user's existing skills.

Based on this comparison, the system identifies the user's **matched skills and missing skills** and calculates the skill match percentage. The results are presented in a clear and structured manner through the Skill Gap Analysis and Career Report modules. The system also generates a **personalized learning roadmap** based on the identified skill gaps, helping users understand which skills they need to develop and follow a suitable learning path. This provides users with a clear understanding of their current skill level and the areas that require improvement.

The system is developed using **Python and Flask** for backend processing, routing, authentication, skill analysis, roadmap generation, and report generation. **HTML and CSS** are used to design the user interface, while **MySQL** is used for storing and managing user and analysis data. By combining career selection, skill gap analysis, personalized roadmap generation, and career reporting in a single system, the proposed application provides a structured approach to career planning and helps users make better decisions about their future skill development.

DECLARATION

I hereby declare that this Dissertation entitled **"SKILL GAP ANALYSIS AND PERSONALIZED ROADAP GENERATOR"** submitted by me for the award of the degree of **BACHELOR OF SCIENCE IN COMPUTER SCIENCE WITH DATA SCIENCE** is the result of my original and independent project work carried out under the guidance of **Dr. S. NARASIMMA SUBRAMANIAN M.C.A., M.Phil., Ph.D.** Assistant Professor & Head of the Department of Computer Science with Artificial Intelligence, Sri Paramakalyani College, Alwarkurichi. I also declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources.

S.No	Name	Register Number
1	AMALA JENIFA A	24081231802912103
2	PESILA V	24081231802912125

Place: Alwarkurichi

Date:

Signature of the Candidate

1.

2.

ACKNOWLEDGEMENT

First and foremost, I would like to thank the Lord Almighty, who bestowed His divine grace abundantly, which carried me through every step of my project work.

I am exclusively thankful to Professor **Dr.C.Muthulakshmi@Sasikala, M.Sc.,M.Phil.,B.Ed.,Ph.D.** the Principal of Sri Paramakalyani College, Alwarkurichi for providing all necessary facilities to carry this project successfully.

I am exclusively thankful to Professor **Dr.V.Ganapathisankarakrishnan, M.Com.,M.Phil.,Ph,D.** the Director of Sri Paramakalyani College, Alwarkurichi for providing all necessary facilities to carry this project successfully.

I express my sincere gratitude to **Dr.T.ARUL RAJ B.Ed., M.Sc.,M.Phil.,Ph.D.** Department of Computer Science, Sri Paramakalyani College, Alwarkurichi.

No words could express my gratitude to my Supervisor **Dr. S. Narasimma Subramanian M.C.A., M.Phil., Ph.D.** Assistant Professor & Head of the Department of Computer Science with Artificial Intelligence, Sri Paramakalyani College, Alwarkurichi for her precious and profound guidance in every part of this work.

I thank all the faculty members and non-teaching staff of the Department of Computer Science, Sri Paramakalyani College, Alwarkurichi for their support and help at every stage of this project work.

I also thank my friends who supported my aspirations with real encouragement and precious suggestions in each and every phase of my project work. I also thank each and every Person who has helped me directly or indirectly for completing this project work.

AMALA JENIFA A

PESILA V

TABLE OF CONTENTS

Chapter No.	Title	Page.No
	ABSTRACT	
	ACKNOWLEDGEMENT	
I	**INTRODUCTION**	1
II	**SYSTEM ANALYSIS**	
	2.1 Existing System	9
	2.2 Limitations of Existing System	11
	2.3 Proposed System	14
III	**SYSTEM SPECIFICATION**	
	3.1 Hardware Specification	20
	3.2 Software Specification	22
	3.3 Software Description	24
IV	**SYSTEM DESIGN**	
	4.1 Database Design	26
	4.2 Dataflow Diagram	30
	4.3 Flow Chart	32
V	**SYSTEMIMPLEMENTATION**	
	5.1 Module Description	35
	5.2 Form Design	48
VI	**SYSTEM TESTING**	53
VII	**CONCLUSION AND FUTURE ENHANCEMENT**	
	7.1 Conclusion	67
	7.2 Future Enhancement	69
	BIBILIOGRAPHY	71

I. INTRODUCTION

1.1 Introduction:

In the present digital era, technology has become an essential part of education, employment, and professional development. The rapid advancement of software technologies has resulted in the continuous creation of new job roles and career opportunities. At the same time, the skills required for existing job roles are also changing regularly. As a result, students, graduates, and job seekers need to continuously improve their knowledge and technical skills to meet the requirements of their desired career. However, many individuals are not fully aware of the skills they already possess, the skills required for a particular career, or the specific areas in which they need improvement.

Choosing a suitable career is an important decision for students and job seekers. A career decision generally depends on several factors, including educational qualification, interests, existing knowledge, technical skills, and future career goals. Without proper guidance, individuals may select a career without understanding its actual skill requirements. This can result in a mismatch between their current capabilities and the expectations associated with the selected career. Identifying this mismatch at an early stage can help individuals plan their learning activities more effectively.

A **skill gap** refers to the difference between the skills currently possessed by an individual and the skills required for a particular career role. For example, a person interested in becoming a Web Developer may already have knowledge of HTML, but may require additional knowledge of CSS, JavaScript, databases, or other relevant technologies. Understanding such differences allows the individual to focus on the areas that require improvement rather than spending unnecessary time learning skills that are already known.

The project **"Skill Gap Analysis & Personalized Roadmap Generator"** is developed to address this requirement through a web-based application. The system provides users with a structured environment where they can create and manage their profile, select a preferred career role, and analyze their current skills against the skills required for that career. The system uses career-related skill information from a dataset to perform the comparison and generate meaningful results.

The application begins with a user authentication process. After successful login, the user can provide relevant profile information such as personal and educational details. The user can then select a preferred career role from the available career options. The selected career role acts as the basis for the subsequent skill analysis process.

Once the career role is selected, the system identifies the skills associated with that role and compares them with the user's current skills. The skills that are present in both sets are treated as matched skills, while the required skills that are not available in the user's current skill set are identified as missing skills. The system also calculates a skill match percentage to provide a simple numerical representation of the user's current alignment with the selected career requirements.

The results of the analysis are presented through a dedicated Skill Gap page. This page allows the user to understand the relationship between their existing skills and the expected skills for the selected career. Instead of providing only a general career recommendation, the system gives specific information about what the user already knows and what needs to be learned.

One of the important features of the proposed system is the **Personalized Roadmap Generator**. The roadmap is generated according to the user's selected career and identified skill gaps. The system focuses on missing skills rather than providing a generic learning sequence. Each roadmap item can provide access to external learning resources related to the particular skill. This approach allows users to follow a more focused learning path based on their individual requirements.

The system also generates a **Career Report** that summarizes the user's analysis results. The report can contain details such as the selected career role, current skills, required skills, matched skills, missing skills, and skill match percentage. This summary helps users review their current position and understand the areas that require further development.

The application is implemented using modern web development technologies. **Python with Flask** is used to develop the backend and handle application routes and processing logic. **MySQL** is used for storing user information, profile details, career selections, and analysis-related data. **HTML and CSS** are used to develop the web pages and provide a user-friendly interface. A career-skill dataset is used to associate different career roles with their required technical skills.

The system follows a modular approach so that each major function can be developed and maintained independently. The major modules include User Login, Profile Management, Career Selection, Skill Gap Analysis, Skill Matching, Personalized Roadmap, and Career Report. These modules are connected to provide a complete flow from user registration/login to skill analysis and personalized learning guidance.

The main purpose of the project is not only to identify a user's skill gap but also to transform the analysis into an actionable learning plan. By connecting skill analysis with a personalized roadmap, the system provides users with a clearer understanding of their learning requirements. This can reduce confusion during career preparation and help users concentrate on relevant skills.

Overall, the Skill Gap Analysis & Personalized Roadmap Generator provides a structured approach to career- oriented skill development. It combines user information, career requirements, skill comparison, and roadmap generation into a single web-based platform. The system demonstrates how software technologies can be used to support individuals in understanding their skill requirements and planning their learning journey in a systematic manner.

1.2 Objectives of the Project:

The primary objective of the Skill Gap Analysis & Personalized Roadmap Generator is to develop a web-based system that helps users understand the difference between their existing skills and the skills required for a selected career role. The system is designed to provide structured career-oriented skill information and generate a personalized learning roadmap based on the identified skill gaps.

The major objectives of the project are described below.

1. To provide a structured career guidance platform

The first objective is to develop a centralized platform through which users can manage their career-related information. Instead of depending on multiple sources to understand career requirements, users can access profile management, career selection, skill analysis, roadmap generation, and reporting features through a single application.

2. To collect and manage user profile information

The system provides a profile management facility through which users can enter and maintain their relevant information. Details such as personal information, educational qualification, and other career-related information can be maintained in the system. This information provides the basic context required for the career planning process.

3. To allow users to select a preferred career role

Different users may have different career interests. Therefore, the system provides a career selection facility that allows users to choose a career role according to their preference. The selected career becomes the basis for identifying the required skills and performing the subsequent skill gap analysis.

4. To identify the skills required for a career

Every career role requires a specific combination of technical skills. One of the objectives of the system is to maintain career-related skill requirements through the available career dataset. When a user selects a career role, the system identifies the corresponding required skills for that role.

5. To analyze the user's existing skills

The system aims to identify and organize the technical skills currently possessed by the user. These skills are used as input for comparison with the skills required for the selected career. This enables the system to determine the user's current skill profile in relation to the desired career.

6. To identify the skill gap

The central objective of the project is to identify the gap between the user's current skills and the required skills for the selected career. The system compares the two skill sets and separates the results into matched skills and missing skills. This provides users with a clear understanding of the areas where additional learning is required.

7. To calculate skill matching percentage

The system calculates a skill match percentage based on the comparison between the user's current skills and the required skills associated with the selected career. This percentage provides a simple numerical representation of the user's current level of skill alignment with the selected career requirements.

8. To generate a personalized learning roadmap

Another major objective is to generate a learning roadmap based on the user's identified missing skills. The roadmap is designed to avoid unnecessary learning recommendations by focusing on skills that are not yet available in the user's current skill set. This makes the learning process more focused and relevant to the selected career.

9. To provide relevant learning resources

The roadmap is intended to connect individual missing skills with suitable external learning resources. For example, if a particular web development skill is identified as missing, the corresponding learning option can direct the user to an external resource related to that skill. This provides a practical starting point for skill development.

10. To generate a comprehensive career report

The system aims to provide a summarized report of the user's career analysis. The report presents important information such as the selected career role, current skills, required skills, matched skills, missing skills, and skill match percentage. This helps users review their results in an organized format.

11. To provide a user-friendly interface

A further objective is to design a simple and visually understandable interface. Users should be able to navigate between the different sections of the system without requiring advanced technical knowledge. The interface therefore includes clearly organized pages for profile management, career selection, skill analysis, roadmap, and report generation.

12. To integrate the complete career preparation process

The project aims to connect multiple stages of career preparation into one continuous workflow. The process begins with user login and profile management, followed by career selection, skill analysis, skill matching, roadmap generation, and career reporting. Integrating these stages makes the system more systematic and easier to use.

13. To reduce confusion in skill development

Many learners are uncertain about which skills they should learn first. By identifying missing skills and presenting them through a roadmap, the system aims to provide clearer direction. Users can focus on the skills associated with their selected career rather than following an unstructured learning process.

14. To demonstrate practical application of web technologies

From a technical perspective, the project aims to demonstrate the practical use of Python, Flask, MySQL, HTML, CSS, and dataset-based processing in the development of a real-world application. It provides an example of how backend processing, database management, user interfaces, and skill comparison logic can be integrated into a single web application.

Therefore, the overall objective of the project is to create a practical and personalized platform that helps users understand their current skills, identify the skills required for a desired career, recognize their skill gaps, and follow a focused learning roadmap for further development.

1.3 Scope of the Project:

The scope of the Skill Gap Analysis & Personalized Roadmap Generator covers the development of a web-based platform for career-oriented skill assessment and personalized learning guidance. The system focuses on helping users understand their current skill position with respect to a selected career role and provides a structured path for improving the identified missing skills.

The system is designed to support the complete flow from user authentication to career analysis and roadmap generation. The scope includes several functional and technical areas.

User Authentication:

The system provides a login mechanism to ensure that users can access their personal career information securely. User authentication helps maintain individual sessions and ensures that profile and analysis information is associated with the appropriate user.

Profile Management:

The project includes a profile management facility where users can provide and update relevant personal and educational information. Maintaining profile information creates a basic user record that can be used throughout the career planning process.

Career Selection:

The system allows users to select a preferred career role. The career selection acts as the primary input for the skill analysis process. Different career roles can be associated with different sets of required technical skills through the career dataset.

Skill Requirement Identification:

For the selected career role, the system identifies the skills required for that particular career. These requirements provide the reference set against which the user's current skills are compared.

Current Skill Analysis:

The scope of the project includes collecting and processing the user's current skill information. These skills are organized and compared with the required skills associated with the selected career.

Skill Gap Identification:

The system determines the difference between the current skill set and the required skill set. Skills that are available in both sets are identified as matched skills, while required skills that are not present in the user's current skills are identified as missing skills.

This functionality provides users with a direct understanding of the areas in which they need additional learning.

Skill Matching:

The project includes a skill matching mechanism that compares current skills with required career skills. The matching result is represented using a match percentage. This provides users with a simple way to understand how closely their current skill set aligns with the selected career requirements.

Personalized Roadmap Generation:

One of the major areas covered by the project is personalized roadmap generation. The roadmap is created based on the missing skills identified during skill gap analysis. Instead of providing an identical roadmap to every user, the system focuses on the individual user's identified learning requirements.

The roadmap can organize missing skills into learning steps and provide a starting point for skill development. This makes the learning process more structured and career-oriented.

Learning Resource Access:

The scope also includes providing external learning resources for roadmap skills. When a user selects a learning option associated with a particular skill, the system can direct the user to a relevant external learning resource. This enables the user to continue learning outside the application.

Career Report Generation:

The system provides a career report that summarizes the user's analysis results. The report can include the selected career role, required skills, current skills, matched skills, missing skills, and skill match percentage. The report gives users a consolidated view of their career skill status.

Database Management:

The project uses MySQL for storing and retrieving application-related information. The database can maintain user account details, profile information, career-related data, and analysis records. Database integration allows information to be retained and accessed during different stages of the application.

User Interface:

The scope of the project also includes the design of a responsive and visually organized web interface. The system provides separate pages for major functions while maintaining consistent navigation and styling. The interface is designed to make career information and analysis results easier to understand.

Technical Scope:

From a development perspective, the project demonstrates the integration of several technologies. Flask manages the backend routes and application logic, Python performs data processing and skill comparison, MySQL provides persistent data storage, and HTML and CSS are used to create the frontend interface. The career-skill dataset provides the reference information required for skill analysis.

Future Expansion:

The current system provides the basic foundation for personalized skill analysis and roadmap generation. Its scope can be extended in the future by adding additional career roles, larger skill datasets, more detailed learning resources, progress tracking, advanced recommendation techniques, and other career development features.

Overall, the scope of the project is centered on transforming career selection and skill assessment into a structured digital process. The system connects user information, career requirements, skill comparison, gap identification, roadmap generation, and reporting within a single platform. This provides a foundation for helping learners understand what they need to learn and organize their career preparation more systematically.

II. SYSTEM ANALYSIS

2.1 Existing System:

Career selection and skill development are important activities for students, graduates, and job seekers. In the existing environment, individuals generally depend on different sources such as general career guidance websites, online learning platforms, search engines, educational resources, social media content, and advice from teachers or professionals to understand career opportunities and skill requirements. Although these resources provide useful information, they are often separated from one another and may not provide a complete picture of an individual's current skill level.

Traditional career guidance methods generally begin with identifying a career based on a person's interests, educational qualification, or general understanding of the job market. Once a career is selected, the individual has to independently search for the skills required for that career. The user then needs to compare those requirements with their own knowledge and skills. This process requires considerable effort because information about career requirements, skill lists, and learning resources is usually available across different platforms.

In many existing systems, career recommendations are provided as general suggestions. A user may receive information about different career roles and their descriptions, but the system may not necessarily analyze the user's current technical skills in detail. As a result, the user may know which career they are interested in but may not clearly understand which skills they already possess and which skills they still need to develop.

Another common approach is the use of online skill assessment platforms. Such systems may provide quizzes, tests, or assessments to measure a user's knowledge in a particular technical area. The results can indicate the user's performance in the selected assessment. However, assessment results alone may not provide a complete career-oriented learning path. The user may still need to identify the relationship between the assessment result and the skills required for a particular career.

Online learning platforms provide another source of career development information. These platforms offer courses and tutorials covering programming languages, web development, databases, data analysis, and other technical areas. However, most learning platforms are designed primarily for course delivery rather than personalized career skill-gap analysis. Users are generally required to decide what they should learn and in what order.

Search engines are also commonly used by students and job seekers to identify career requirements. A user may search for phrases such as “skills required for a Web Developer” or “skills required for a Data Analyst.” The search results may contain multiple articles, videos, courses, and websites. Although this provides access to a large amount of information, the user must manually evaluate and organize the information.

In the existing approach, there is often no single integrated workflow connecting user profile information, career selection, skill analysis, skill matching, missing-skill identification, and personalized roadmap generation. These activities are usually performed separately. The user has to move between different platforms to complete the process.

For example, consider a student who wants to become a Web Developer. The student may first search for the skills required for web development. The student may then prepare a list of technologies such as HTML, CSS, JavaScript, database concepts, and other relevant skills. After that, the student must compare this list with the skills they already know. If the student identifies missing skills, another search is required to find appropriate learning resources. Finally, the student must create a learning sequence manually.

This process can be time-consuming and may lead to an unstructured learning approach. Different websites may also present different skill lists depending on their source and purpose. Therefore, users may find it difficult to determine which information is relevant to their specific career goal.

Existing career guidance approaches can therefore be described as information-oriented rather than fully personalized. They provide career descriptions, skill information, assessments, or learning resources, but the user is often responsible for connecting these elements together.

The existing system also generally does not maintain a continuous relationship between the user's selected career and the identified skill gap. If the user changes their career preference, they may need to repeat the entire information-gathering and comparison process manually.

Another challenge is the lack of a consolidated career report. Users may collect information from different websites, but this information is not necessarily organized into a single summary containing their current skills, required skills, matched skills, missing skills, and overall skill matching percentage.

Therefore, while existing career guidance and online learning systems provide valuable resources, there is a need for an integrated platform that can combine career selection, skill analysis, skill matching, personalized roadmap generation, and career reporting in a single system.

The proposed Skill Gap Analysis & Personalized Roadmap Generator addresses this requirement by bringing these functions together within one web-based application. The system uses the selected career role and the user's current skills as the basis for generating a personalized skill analysis and learning roadmap.

2.2 Limitations of Existing System:

Although existing career guidance systems, online learning platforms, and skill assessment websites provide useful information, several limitations can be identified when they are considered from the perspective of personalized career preparation.

1. Lack of Personalized Skill Analysis

One of the major limitations is that general career guidance platforms may not analyze an individual's current skills against the requirements of a selected career. They may provide a general list of skills but do not always determine which of those skills are already known by the user.

As a result, the user has to perform the comparison manually. This can make it difficult for beginners to understand their exact skill gap.

2. Generalized Career Recommendations

Many existing systems provide general career information based on broad categories such as educational qualification, interests, or job descriptions. Such information may not consider the user's complete current skill profile.

A career role may be suitable from a general perspective, but the user may still require several additional skills before being prepared for that role. Without skill-level comparison, the user may not clearly understand this requirement.

3. Manual Skill Comparison

In traditional methods, users often have to collect the required skills for a career and compare them manually with their own skills. This process becomes difficult when the number of skills increases.

Manual comparison can also result in incomplete identification of missing skills. A user may accidentally ignore a required skill or consider a skill equivalent when the system would distinguish them separately.

4. Scattered Information

Career descriptions, skill requirements, learning materials, and assessment results are generally available on different websites or platforms. Users have to search multiple sources to gather all the required information.

This creates a fragmented learning experience. Instead of following one continuous workflow, users have to repeatedly switch between different sources.

5. No Direct Skill Gap to Roadmap Connection

Another limitation is the lack of direct connection between skill-gap identification and learning roadmap generation. Some systems can provide skill information, while other platforms provide learning resources, but the user is responsible for deciding what to learn first.

Without a personalized roadmap, users may spend time learning skills that are not immediately relevant to their selected career.

6. Difficulty in Identifying Learning Priorities

When users identify multiple missing skills, they may not know which skill should be learned first. General learning platforms may provide many courses without considering the user's specific career objective.

This can result in an unstructured learning process. A user may begin learning advanced topics before developing the necessary foundational skills.

7. Lack of Consolidated Career Report

Existing approaches may not provide a single report containing all relevant career analysis information. Users may have to maintain their own notes about their current skills, required skills, learning requirements, and progress.

A consolidated report would make it easier for users to review their career preparation status.

8. Limited Integration

Career guidance, skill assessment, learning resources, and reporting are often separate functionalities. The lack of integration increases the amount of effort required from users.

For example, selecting a career may happen on one platform, skill assessment on another platform, and learning on a third platform. There is generally no continuous flow connecting these activities.

9. Repetition of User Effort

When users use multiple independent platforms, they may have to enter similar information repeatedly. This increases the effort required and may discourage users from completing a detailed career analysis.

10. Difficulty for Beginners

Users with limited knowledge about technology may find it difficult to determine the relevance of different skills. A beginner may encounter a large number of technical terms without understanding which ones are essential for the selected career.

A structured skill-gap system can reduce this difficulty by presenting information according to the user's selected career.

11. No Career-Specific Learning Focus

General learning platforms are designed to support a broad range of learners. Therefore, they may provide many courses without directly connecting them to one particular career role.

Users need to determine which courses are relevant to their career objective. This creates additional decision- making effort.

12. Limited Use of User-Specific Data

Some systems do not continuously use the user's profile and selected career information to personalize subsequent results. Without user-specific processing, the recommendations may remain general.

13. Difficulty in Tracking the Current Skill Position

Without a dedicated analysis and report mechanism, users may find it difficult to understand their current position after learning additional skills. A structured system can provide a clearer comparison between current skills and career requirements.

14. Lack of an Integrated Learning Direction

The biggest limitation of the existing approach is the absence of a unified process that starts with the user's profile and career preference and ends with a personalized learning roadmap. Users are generally required to connect each stage themselves.

The identified limitations demonstrate the need for an integrated system that can perform career-oriented skill comparison and provide a structured learning direction. The proposed system is designed to address these limitations by combining profile management, career selection, skill gap analysis, skill matching, roadmap generation, and career reporting in one application.

2.3 Proposed System:

The proposed system is a web-based application titled **"Skill Gap Analysis & Personalized Roadmap Generator."** It is designed to provide a structured and personalized approach to career skill development. The system integrates user profile management, career selection, skill analysis, skill matching, missing-skill identification, personalized roadmap generation, and career reporting into a single platform.

The main concept of the proposed system is to compare the user's current technical skills with the skills required for a selected career role. Based on this comparison, the system identifies the skills that are already available and the skills that need to be developed. The identified missing skills are then used to generate a personalized learning roadmap.

The system begins with a user login process. Each user accesses the application through an authenticated session. This ensures that the user's profile, career preference, and analysis information are associated with the correct account.

After login, the user can access the profile management section. The profile page allows the user to enter and maintain relevant personal and educational details. This information provides the basic user context for the career planning process.

The next stage is career selection. The user can select a preferred career role from the available career options. Each career role is associated with a set of required technical skills maintained through the career dataset. The selected career becomes the reference point for the skill analysis process.

Once the career role has been selected, the system retrieves the corresponding required skills. The user's current skills are then compared with these required skills. This comparison forms the core of the Skill Gap Analysis module.

The skill analysis process separates the results into different categories. Skills that are present in both the user's current skill set and the required career skill set are identified as **matched skills**. Required skills that are not present in the user's current skill set are identified as **missing skills**.

The system also calculates a **skill match percentage**. This percentage represents the proportion of required skills that are matched by the user's current skills. It provides a simple numerical representation of the user's current alignment with the selected career requirements.

For example, if a selected career requires eight skills and the user possesses three of those required skills, the system can identify those three as matched skills and the remaining five as missing skills. The resulting match percentage can be calculated based on the relationship between matched skills and total required skills.

The proposed system then uses the identified missing skills to generate a personalized roadmap. The roadmap is not based on a fixed list of topics for every user. Instead, it is generated according to the current skill gap identified for the selected career. Therefore, different users can receive different roadmap requirements depending on their current skills.

Each roadmap skill can be associated with an external learning resource. The **Start Learning** option can direct users to an appropriate learning resource related to that skill. This creates a connection between skill-gap identification and actual learning activities.

The system also provides a Career Report page. The report summarizes the analysis results in an organized format. It can display the selected career role, current skills, required skills, matched skills, missing skills, and skill match percentage. This enables users to review their career preparation status from a single page.

The proposed application uses **Flask** as the backend framework. Flask manages application routes, session handling, request processing, and communication between the user interface and application logic. **Python** is used for implementing the skill comparison and processing operations.

MySQL is used as the database management system. User-related information, profile information, career- related data, and analysis records can be stored and retrieved through the database. This provides persistent storage for the application.

The frontend is developed using **HTML and CSS**. HTML provides the structure of the application pages, while CSS is used to create the visual appearance, layout, navigation, cards, buttons, and other interface elements. The system uses a consistent navigation structure so that users can move between the major pages easily.

The proposed system follows a modular design. The major modules include:

- User Login Module
- Profile Management Module
- Career Selection Module
- Skill Gap Analysis Module
- Skill Matching Module
- Personalized Roadmap Module
- Career Report Module

These modules are connected to create a continuous workflow. The user first logs in, maintains their profile, selects a career, analyzes their skills, views the identified skill gap, follows the personalized roadmap, and reviews the generated career report.

The proposed system therefore transforms career preparation from a collection of separate activities into an integrated process. It does not simply provide a list of careers or courses; instead, it connects the user's selected career with their current skills and identifies the specific skills that require development.

The system is designed to be simple enough for students and beginners while providing useful technical information for career preparation. Its modular structure also makes it possible to expand the system in the future by adding more career roles, skills, learning resources, progress tracking, and advanced recommendation features.

Overall, the proposed system provides a practical solution for identifying career-related skill gaps and converting those gaps into a personalized learning roadmap. By integrating analysis and learning guidance within one platform, the system provides a more organized approach to career-oriented skill development.

2.4 Advantages of Proposed System:

The proposed Skill Gap Analysis & Personalized Roadmap Generator provides several advantages by integrating career selection, skill comparison, and personalized learning guidance into a single web-based application.

1. Personalized Skill Analysis

The system analyzes the user's current skills in relation to the selected career role. This makes the analysis more specific to the individual rather than providing only general career information.

2. Clear Identification of Skill Gaps

The system separates skills into matched and missing categories. This makes it easier for users to understand what they already know and what they need to learn.

3. Career-Oriented Skill Matching

The skill comparison is performed based on the requirements of the selected career role. Therefore, the analysis is directly connected to the user's career objective.

4. Personalized Learning Roadmap

The roadmap is generated using the identified missing skills. This allows users to focus their learning activities on areas that require improvement.

5. Reduced Manual Effort

The system automatically performs the comparison between current skills and required skills. Users do not have to manually compare large lists of skills.

6. Integrated Platform

Profile management, career selection, skill analysis, skill matching, roadmap generation, and career reporting are available within one application. This reduces the need to use multiple independent platforms for basic career planning activities.

7. Relevant Learning Resources

The roadmap can provide external learning resources for individual skills. This creates a direct connection between identifying a missing skill and finding a resource to learn it.

8. Easy Understanding of Current Skill Position

The skill match percentage provides a simple numerical representation of the user's current alignment with the selected career requirements. Along with matched and missing skills, this gives the user a clearer view of their current status.

9. Organized Career Report

The Career Report module consolidates important analysis information into a single page. Users can review their selected career, current skills, required skills, matched skills, missing skills, and match percentage in an organized manner.

10. User-Friendly Interface

The system provides separate pages for different activities and maintains consistent navigation. This helps users access the required functionality without unnecessary complexity.

11. Centralized Data Management

The use of MySQL provides a structured method for storing user and application information. Data can be retrieved when required by different modules.

12. Modular Architecture

The system is divided into independent functional modules. This makes the application easier to understand, maintain, test, and extend.

13. Suitable for Students and Job Seekers

The system can be useful for individuals who are beginning their career preparation and need structured information about required skills. It provides a simple process for understanding skill requirements and identifying learning areas.

14. Better Learning Focus

Instead of presenting a large number of unrelated learning topics, the roadmap focuses on the skills identified as missing for the selected career. This can help users organize their learning activities more systematically.

15. Expandable System

The system can be extended with additional career roles, larger skill datasets, learning resources, progress tracking, and other features. The modular design provides a foundation for future development.

16. Practical Application of Technology

The project demonstrates how Python, Flask, MySQL, HTML, CSS, and dataset-based processing can be combined to develop a real-world career guidance application.

17. Continuous Career Planning Workflow

The system creates a continuous workflow from login to report generation. Users can move through profile management, career selection, skill analysis, roadmap generation, and reporting without having to manually connect separate systems.

18. Supports Structured Career Preparation

The overall advantage of the system is that it converts a broad career objective into a more structured skill- development process. Users can identify their current skills, understand the required skills, recognize the missing areas, and access learning resources related to those areas.

Thus, the proposed system provides an integrated framework for career-oriented skill analysis and learning guidance. Its primary strength lies in connecting **career selection → skill comparison → skill gap**

identification → personalized roadmap → career report within a single application.

III. SYSTEM SPECIFICATION

3.1 Hardware Specification:

Hardware specification defines the minimum physical resources required to develop, run, test, and maintain the **Skill Gap Analysis & Personalized Roadmap Generator** web application. Since the proposed system is a web-based application developed using Python, Flask, MySQL, HTML, and CSS, it does not require specialized or high-end hardware. A standard personal computer or laptop with adequate processing power, memory, and storage is sufficient for the development and execution of the system.

The hardware requirements are selected based on the needs of the application development environment, database operations, web server execution, and testing activities.

3.1.1 Processor

A modern multi-core processor is recommended for the development and execution of the system. A processor equivalent to an **Intel Core i3 or above** is sufficient for running Python, Flask, MySQL, Visual Studio Code, and a web browser simultaneously.

The processor handles the execution of the Flask application, Python-based skill analysis operations, database queries, and frontend testing. A higher-performance processor can improve response time when multiple development tools and services are running at the same time.

3.1.2 Random Access Memory

A minimum of **4 GB RAM** can be used for basic development and testing, while **8 GB RAM or above** is recommended for smoother performance.

RAM is used by the operating system, code editor, Python runtime, Flask development server, MySQL server, web browser, and other development tools. Adequate memory allows these components to operate simultaneously without excessive system slowdown.

3.1.3 Storage

The project requires sufficient storage for the operating system, source code, database files, datasets, development tools, and supporting resources. A minimum of **256 GB storage** is recommended.

An SSD is preferable because it provides faster file access and improves the startup time of development applications, database services, and the project environment.

3.1.4 Display

A standard monitor or laptop display with a minimum resolution of **1366 × 768 pixels** is sufficient for application development and testing. A higher-resolution display can provide a better development experience, especially when working with multiple code files and browser windows.

3.1.5 Keyboard and Mouse

A standard keyboard and mouse are required for entering source code, configuring the development environment, managing files, interacting with the database, and testing the web application.

3.1.6 Network Connectivity

Internet connectivity is useful during development for installing Python packages, accessing documentation, testing external learning resources, and opening learning-resource links provided through the personalized roadmap.

Although the core application can operate locally, internet access is beneficial when users need to access external learning resources associated with roadmap skills.

3.1.7 Hardware Requirement Summary

The recommended hardware configuration is given below:

Hardware Component	Minimum Requirement
Processor	Intel Core i3 or equivalent
RAM	4 GB minimum
Storage	256 GB
Display	1366 × 768 or above
Keyboard	Standard keyboard
Mouse	Standard mouse
Network	Internet connection recommended

The hardware requirements of the proposed system are therefore relatively moderate. The application is primarily a software-based web application and does not require dedicated graphics processing units or specialized hardware. A standard computer system can support the development, execution, and testing of the complete application.

3.2 SOFTWARE SPECIFICATION

The Software Specification describes the software environment required to develop, execute, and maintain the SKILL GAP ANALYSIS & PERSONALIZED ROADMAP GENERATOR. The project is developed as a web-based application using Python and Flask for the backend, HTML and CSS for the user interface, and MySQL for database management. A career-skill dataset is used to identify the skills associated with different career roles.

The software requirements are selected based on the functionality of the application and the development environment. The following software components are required for the successful execution of the system.

Operating System

The application can be developed and executed in a Windows-based environment. The operating system provides the necessary platform for installing Python, Flask, MySQL, Visual Studio Code, and other supporting software required for development and testing.

Python

Python is the primary programming language used for developing the application. It is used to implement the application logic, process user information, perform skill analysis, compare current skills with required career skills, generate personalized roadmaps, and prepare career reports.

Python provides a simple and flexible programming environment and supports various libraries required for web application development and data processing.

Flask

Flask is used as the web framework for developing the application. It handles URL routing, page rendering, form submission, session management, user authentication, and communication between the user interface and backend logic.

The Flask framework connects the different modules of the application and provides the overall application structure.

MySQL

MySQL is used as the relational database management system for storing and retrieving application data. User account information, profile details, career preferences, and skill analysis-related information can be managed through the database.

The Flask application communicates with the MySQL database to perform required database operations.

HTML

HTML is used to create the structure of the web pages. It defines elements such as headings, forms, buttons, navigation sections, tables, skill information, roadmap sections, and report content.

The HTML templates provide the basic structure through which users interact with the application.

CSS

CSS is used to design and style the web interface. It controls the appearance, layout, spacing, typography, colors, backgrounds, cards, buttons, navigation sidebar, and other visual elements of the application.

Pandas

Pandas can be used for handling and processing structured career and skill data. It supports operations such as reading datasets, filtering career-related information, processing skill lists, and preparing data for skill analysis.

Career-Skill Dataset

A career-skill dataset is used as a source of information for mapping career roles with their required skills. The dataset supports the Skill Gap Analysis and Skill Matching modules by providing the required skill information associated with the selected career.

Visual Studio Code

Visual Studio Code is used as the development environment for writing, editing, organizing, and debugging the project files. It supports Python development and provides a convenient environment for managing Flask application files, HTML templates, CSS files, and other project resources.

Web Browser

A modern web browser such as Google Chrome can be used to access and test the web application. The browser displays the HTML and CSS interface and allows users to interact with forms, navigation links, buttons, roadmap resources, and report pages.

Software Component	Requirement / Purpose
Operating System	Windows
Programming Language	Python
Web Framework	Flask
Database	MySQL
Front-End	HTML, CSS
Data Processing	Pandas
Dataset	Career-Skill Dataset
Development Environment	Visual Studio Code

3.3 Software Description:

The Skill Gap Analysis & Personalized Roadmap Generator is developed as a web-based application using a combination of backend, frontend, database, and data-processing technologies. Each technology has a specific role in the overall system, and their integration enables the application to provide personalized career skill analysis.

3.3.1 Career-Skill Dataset

The career-skill dataset is an important data source for the skill analysis process. It contains information connecting career roles with their required skills.

When the user selects a career role, the application retrieves the corresponding required skills from the available data. These skills form the reference set used during skill comparison.

For example, a career such as Web Developer may have a set of technical requirements associated with it. The system compares these requirements against the user's current skills to determine matched and missing skills.

3.3.2 Skill Analysis Processing

The skill analysis component processes the user's current skill information and compares it with the required skills for the selected career.

The general processing flow can be represented as:

User Current Skills → Career Required Skills → Skill Comparison → Matched Skills + Missing Skills → Match Percentage

This processing provides the foundation for both the Skill Gap page and the Personalized Roadmap page.

3.3.3 Skill Matching

The skill matching process determines which required skills are already available in the user's current skill set.

The matching results are used to identify the skills that the user has already developed. The system also determines the skills that are required but not currently available.

The match percentage provides a quantitative representation of the comparison.

A general calculation can be represented as:

Skill Match Percentage = (Number of Matched Skills / Number of Required Skills) × 100

The calculated percentage is presented to the user along with the detailed skill categories.

3.3.4 Personalized Roadmap Generation

The Personalized Roadmap component uses the missing skills identified during the analysis process.

Instead of generating the same learning plan for all users, the system considers the user's current skill gap. The missing skills are converted into roadmap items that represent areas for further learning.

Learning resources can be associated with roadmap items. The user can select the **Start Learning** option to access an external learning resource relevant to the selected skill.

3.3.5 Career Report

The Career Report component presents a summarized view of the user's analysis. The report can include:

- Selected Career Role
- Current Skills Required
- Skills Matched
- Skills Missing Skills
- Skill Match Percentage

3.3.6 Integration of Software Components

All the software components work together as an integrated system. The frontend sends user inputs to the Flask backend. Flask processes the request using Python logic and communicates with MySQL whenever database information is required.

The career-skill dataset provides the required career reference information. The skill analysis logic compares the user's current skills with the required skills. The resulting information is then passed back to the HTML templates and displayed through the CSS-designed interface.

The overall software interaction can be represented as:

User Interface → Flask Backend → Python Processing → MySQL / Dataset → Skill Analysis → Roadmap / Report → User Interface

This integrated software environment provides the technical foundation required to implement the Skill Gap Analysis & Personalized Roadmap Generator successfully.

IV. SYSTEM DESIGN

4.1 Database Design:

Database design defines the structure used to store and manage information required by the **Skill Gap Analysis & Personalized Roadmap Generator**. MySQL is used as the database management system because it provides structured relational storage and supports efficient insertion, retrieval, modification, and management of application data.

The database is designed to support user authentication, profile management, career selection, skill analysis, and report generation. The stored information allows the application to retrieve the user's previous information whenever required.

4.1.1 User Data

The user-related information is maintained to support authentication and personalized access. Typical user information includes:

- User ID
- Username
- Password
- Email
- Mobile Number

The User ID acts as a unique identifier for each user. It allows other user-related information to be associated with the correct account.

Passwords are handled through secure password processing rather than being stored as plain text. During login, the entered credentials are verified against the stored account information.

4.1.2 Profile Data

The profile information stores additional details entered by the user after login.

The profile can contain information such as:

- User ID
- Full Name
- Category
- Educational Qualification
- Other career-related information

The User ID provides the relationship between the account and its corresponding profile.

4.1.3 Career Data

Career information is used to identify the available career roles and their associated requirements.

The career-related data can contain:

- Career Role
- Required Skills
- Career Description or related information

The required skills act as the reference information for the skill-gap calculation.

4.1.4 Analysis Data

The analysis data stores the results generated after comparing the user's current skills with the required skills of the selected career.

The analysis information can include:

- User ID
- Career Role
- Current Skills
- Required Skills
- Matched Skills
- Missing Skills
- Match Percentage
- Analysis Creation Time

The User ID connects the analysis with the appropriate user account.

4.1.5 Relationship Between Data

The different data components are logically connected through the User ID and selected Career Role.

A simplified relationship can be represented as:

User → Profile

User → Career Selection

User → Skill Analysis

Career Role → Required Skills

Skill Analysis → Matched Skills + Missing Skills

Missing Skills → Personalized Roadmap

This structure enables the application to retrieve information according to the currently logged-in user.

4.1.6 Database Operations

The system performs several database operations during its execution.

INSERT: Used when new user, profile, career, or analysis information needs to be stored.

SELECT: Used to retrieve user information, profile details, career requirements, and analysis results.

UPDATE: Used when existing profile or career-related information is modified.

DELETE: Can be used when records need to be removed according to application requirements.

4.1.7 Data Flow Through Database

When the user logs in, the application retrieves the corresponding account information from MySQL. After authentication, the user's ID is maintained through the session.

When profile information is submitted, it is stored in the profile-related database structure. When a career is selected, the selected career information is associated with the user.

During skill analysis, the system retrieves the user's current skills and the required skills for the selected career. After performing the comparison, the resulting analysis can be stored for later retrieval.

The Report page can then retrieve the latest analysis record and display the stored information to the user.

4.1.8 Database Design Benefits

The database design provides several benefits:

1. Organized storage of application data.
2. Easy retrieval of user-specific information.
3. Reduced duplication of information.
4. Persistent storage of analysis results.
5. Easy integration with Flask.
6. Support for multiple users.
7. Better management of career and skill information.
8. Ability to maintain analysis history.

The database therefore acts as an important foundation of the application. It ensures that user information and analysis results are stored systematically and can be retrieved whenever required by the different modules.

Database Tables

```
mysql> use skillnavigator;
Database changed
mysql> show tables;
+--------------------------+
| Tables_in_skillnavigator |
+--------------------------+
| analysis                 |
| career_preferences       |
| career_skills            |
| learning_progress        |
| profile                  |
| users                    |
+--------------------------+
6 rows in set (0.02 sec)
```

Analysis Table

```
mysql> desc analysis;
+------------------+--------------+------+-----+-------------------+-------------------+
| Field            | Type         | Null | Key | Default           | Extra             |
+------------------+--------------+------+-----+-------------------+-------------------+
| id               | int          | NO   | PRI | NULL              | auto_increment    |
| user_id          | int          | NO   |     | NULL              |                   |
| career_role      | varchar(255) | NO   |     | NULL              |                   |
| current_skills   | text         | YES  |     | NULL              |                   |
| required_skills  | text         | YES  |     | NULL              |                   |
| matched_skills   | text         | YES  |     | NULL              |                   |
| missing_skills   | text         | YES  |     | NULL              |                   |
| total_required   | int          | YES  |     | 0                 |                   |
| total_matched    | int          | YES  |     | 0                 |                   |
| total_missing    | int          | YES  |     | 0                 |                   |
| skill_percentage | decimal(5,2) | YES  |     | 0.00              |                   |
| analysis_type    | varchar(50)  | YES  |     | NULL              |                   |
| created_at       | timestamp    | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
| match_percentage | float        | YES  |     | 0                 |                   |
+------------------+--------------+------+-----+-------------------+-------------------+
14 rows in set (0.01 sec)
```

Career_preferences

```
mysql> desc career_preferences;
+----------------+--------------+------+-----+-------------------+-------------------+
| Field          | Type         | Null | Key | Default           | Extra             |
+----------------+--------------+------+-----+-------------------+-------------------+
| id             | int          | NO   | PRI | NULL              | auto_increment    |
| user_id        | int          | NO   |     | NULL              |                   |
| full_name      | varchar(100) | YES  |     | NULL              |                   |
| career_role    | varchar(150) | NO   |     | NULL              |                   |
| current_skills | text         | YES  |     | NULL              |                   |
| created_at     | timestamp    | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+----------------+--------------+------+-----+-------------------+-------------------+
6 rows in set (0.00 sec)
```

4.2 Data Flow Diagram:

A Data Flow Diagram (DFD) represents the movement of data between users, application processes, databases, and other components of the system. It provides a graphical representation of how input data is received, processed, stored, and returned as output.

The Data Flow Diagram for the **Skill Gap Analysis & Personalized Roadmap Generator** can be represented using different levels.

4.2.1 Level 0 DFD

At Level 0, the complete application can be represented as a single main process.

The main external entity is the **User**.
The user provides:

- Login credentials
- Profile information
- Career preference
- Current skill information

The system processes the information and provides:

- Skill Gap Analysis
- Skill Matching Results
- Personalized Roadmap
- Career Report

The basic Level 0 flow is:

User → Skill Gap Analysis & Personalized Roadmap Generator → User

4.2.2 Level 1 DFD

At Level 1, the main system is divided into several major processes.

Process 1 – User Authentication

The user provides username and password. The system verifies the credentials using the user database.

User → Login Process → User Database

The result is successful authentication or an appropriate error response.

Process 2 – Profile Management

The user enters profile information. The system validates and stores the information in the profile database.

User → Profile Management → Profile Database

Process 3 – Career Selection

The user selects a preferred career role. The system stores the selected career and retrieves the corresponding required skills.

User → Career Selection → Career-Skill Data

Process 4 – Skill Gap Analysis

The system receives the user's current skills and compares them with the required skills for the selected career.

Current Skills + Required Skills → Skill Gap Analysis

The output consists of matched skills and missing skills.

Process 5 – Skill Matching

The system calculates the relationship between matched skills and total required skills.

Matched Skills + Required Skills → Skill Matching → Match Percentage

Process 6 – Personalized Roadmap

The missing skills are passed to the roadmap generation process.

Missing Skills → Roadmap Generation → Personalized Learning Roadmap

Process 7 – Career Report

The analysis information is collected and presented as a consolidated report.

Analysis Data → Report Generation → Career Report

4.2.3 Data Stores

The major data stores involved in the system include:

- User Database
- Profile Database
- Career-Skill Dataset
- Analysis Database

These data stores provide the information required by the different processes.

4.2.4 Overall Data Flow

The complete data flow can be summarized as:

User Login → Profile Information → Career Selection → Required Skills → Current Skills → Skill Comparison → Matched/Missing Skills → Match Percentage → Personalized Roadmap → Career Report

4.3 System Flowchart:

A System Flowchart represents the sequential flow of operations performed by the **Skill Gap Analysis & Personalized Roadmap Generator**. It describes how the system processes user information from login through skill analysis, roadmap generation, and report presentation.

Step 1 – Start

The application begins when the user opens the web application.

Step 2 – User Login

The user enters their username and password.

The system checks whether the credentials are valid.

If the credentials are invalid, an error message is displayed and the user can attempt login again. If the credentials are valid, the user is redirected to the dashboard.

Step 3 – Dashboard

After successful login, the user accesses the dashboard. From the dashboard, the user can navigate to profile, career selection, skill analysis, roadmap, and report sections.

Step 4 – Profile Management

The user enters or updates the required profile information.

The system stores the profile information in the database.

Step 5 – Career Selection

The user selects a preferred career role.

The system identifies the required skills associated with that career.

Step 6 – Current Skill Input

The user's current skills are obtained for the skill analysis process.

Step 7 – Skill Comparison

The system compares the user's current skills with the required skills of the selected career.

The comparison produces two major categories:

Matched Skills

and

Missing Skills

Step 8 – Match Percentage

The system calculates the skill match percentage based on the number of matched skills compared with the total number of required skills.

Step 9 – Display Skill Gap

The system displays the skill analysis results to the user. The user can view current skills, required skills, matched skills, missing skills, and match percentage.

Step 10 – Generate Personalized Roadmap

The missing skills identified during the analysis are used to generate the personalized roadmap.

The roadmap presents learning steps corresponding to the user's skill gap.

Step 11 – Learning Resource Access

The user can select the Start Learning option for a roadmap skill. The system provides access to an external learning resource associated with that skill.

Step 12 – Generate Career Report

The system collects the analysis information and presents it through the Career Report page.

Step 13 – Logout

After completing the required activities, the user can logout from the system. The session is terminated.

Overall Flow

The complete flow can be summarized as:

START

↓

LOGIN

↓

Validate Credentials

↓

Valid?

→ No → Display Error → Login Again

→ Yes

↓

Dashboard

Dashboard

↓

Manage Profile

↓

Select Career

↓

Identify Required Skills

↓

Obtain Current Skills

↓

Compare Skills

↓

Matched Skills + Missing Skills

↓

Calculate Match Percentage

↓

Display Skill Gap

↓

Generate Personalized Roadmap

↓

Access Learning Resources

↓

Generate Career Report

↓

Logout

↓

END

The system flowchart provides a clear representation of the complete operational sequence of the application. It helps developers understand the logical flow and also provides users and evaluators with a simple overview of how the system functions from beginning to end.

V. SYSTEM IMPLEMENTATION

5.1 Module Description

The **Skill Gap Analysis & Personalized Roadmap Generator** is implemented as a modular web-based application. The system is divided into several functional modules, where each module performs a specific task within the overall career planning process. The modular approach makes the system easier to develop, test, maintain, and extend.

The major modules implemented in the system are:

1. User Login Module
2. Profile Management Module
3. Career Selection Module
4. Skill Gap Analysis Module
5. Skill Matching Module
6. Personalized Roadmap Module
7. Career Report Module

These modules are interconnected and follow a sequential workflow. The user first logs into the application, manages their profile, selects a career, and provides their current skill information. The system then performs skill gap analysis and skill matching. Based on the identified missing skills, a personalized roadmap is generated. Finally, the user can view a consolidated career report.

The backend of the system is implemented using Python and Flask, while MySQL is used for database management. HTML and CSS are used to develop the user interface. The career-skill dataset provides the required skill information associated with different career roles.

5.1.1 User Login Module

The User Login Module is the entry point of the Skill Gap Analysis & Personalized Roadmap Generator. It is responsible for authenticating users before allowing them to access their personalized career information.

The main purpose of this module is to ensure that the user's profile and career-related information are accessed only through the appropriate authenticated session.

User Input

The login interface collects the following information:

Username
Password

The user enters these credentials through the login form and submits the request to the Flask backend.

Authentication Process

When the login form is submitted, the Flask application receives the username and password. The backend searches the MySQL database for the corresponding user account.

The stored password is not treated as ordinary plain-text information. Password verification is performed using secure password hashing and comparison techniques.

If the entered credentials match the stored account information, the login process is considered successful.

Session Management

After successful authentication, the application creates a user session. Information associated with the logged-in user can be maintained through the Flask session mechanism.

The session allows different pages of the application to identify the currently logged-in user. This is important because profile information, career selection, skill analysis, roadmap information, and reports must be associated with the correct user.

Invalid Login

If the username or password is incorrect, the system does not allow access to protected pages. Instead, the user receives an appropriate error message and can attempt to login again.

Security

Authentication provides a basic security layer for the application. Routes that contain user-specific information check whether a valid user session exists before displaying the requested page.

Output

After successful login, the user is redirected to the Dashboard. From there, the user can access the other modules of the application.

The overall process can be represented as:

Username + Password → Database Verification → Authentication → Session Creation → Dashboard

Thus, the User Login Module provides controlled access to the personalized features of the application.

5.1.2 Profile Management Module:

The Profile Management Module allows users to enter and maintain their personal and educational information. It is an important part of the application because the career planning process is associated with an individual user's information.

After successful login, the user can access the Profile page through the application navigation.

Profile Information

The module can collect information such as:

Full Name
Category
Educational Qualification
Other relevant profile information

The information entered by the user is submitted through an HTML form.

Data Processing

When the profile form is submitted, the Flask backend receives the form data. The application validates the information and stores or updates it in the MySQL database.

The user's unique ID is used to associate the profile information with the appropriate account.

Profile Update

The module allows users to modify their profile information when required. If the user changes their educational qualification or other details, the updated information is stored in the database.

This prevents users from having to create a new account whenever profile information changes.

Integration with Other Modules

The profile module works together with the career selection and dashboard components. After profile information is saved successfully, the application can redirect the user to the next stage of the career planning process.

The profile data also provides additional context for the overall career planning workflow.

User Interface

The Profile page provides clearly organized input fields and a Save option. Validation and feedback messages can be displayed when information is successfully stored or when an input requires correction.

Output

After successful profile submission, the user receives confirmation that the profile has been saved. The stored information can later be retrieved and displayed when the user accesses the Profile page again.

The process can be represented as:

Profile Form → Flask Backend → Validation → MySQL Database → Updated Profile

The Profile Management Module therefore provides a structured method for maintaining user information and forms an important foundation for personalized career planning.

5.1.3 Career Selection Module:

The Career Selection Module allows users to specify the career role they are interested in pursuing. The selected career plays a central role in the remaining stages of the system because the required skills for the selected career are used as the reference for skill gap analysis.

Career Selection

After completing or updating their profile, the user can access the Career Selection page. The system provides available career options from the career-related data.

The user selects the career role that matches their current career interest.

Career-Skill Relationship

Each career role is associated with a set of required technical skills. These skill requirements are obtained from the career-skill dataset used by the application.

For example, a Web Developer career may be associated with several web development-related technical skills. The exact skill requirements depend on the career data maintained by the system.

Processing

When the user submits the selected career, the Flask backend receives the career preference and associates it with the user's account.

The selected career becomes the input for the Skill Gap Analysis Module.

Navigation

After the career preference is saved successfully, the system can redirect the user to the Skill Gap Analysis page. A confirmation message can also be displayed to indicate that the career preference has been saved.

Integration

The Career Selection Module provides the connection between the user's career goal and the skill analysis process.

The overall relationship is:

User Profile → Career Selection → Required Career Skills → Skill Gap Analysis

Output

The selected career role is displayed on subsequent pages such as the Skill Gap page, Roadmap page, and Career Report page.

This ensures that the user's analysis and learning roadmap remain connected to the selected career objective.

The Career Selection Module therefore provides the starting point for personalized skill comparison and roadmap generation.

5.1.4 Skill Gap Analysis Module:

The Skill Gap Analysis Module is the core component of the proposed system. Its main purpose is to identify the difference between the user's current technical skills and the skills required for the selected career role.

The module uses two primary inputs:

1. User's Current Skills
2. Required Skills for the Selected Career

The comparison between these two sets produces the user's skill gap.

Current Skills

The user's current skills represent the technical knowledge already possessed by the individual. These skills are used as the user's current skill set during the analysis.

Required Skills

The required skills are obtained according to the career role selected by the user. The career-skill dataset provides the reference list of skills associated with the selected career.

Skill Comparison

The system compares the current skill set with the required skill set.

If a skill exists in both sets, it is identified as a **Matched Skill**.

If a skill is required for the selected career but is not present in the user's current skill set, it is identified as a **Missing Skill**.

The general process is:

Current Skills + Required Skills → Comparison → Matched Skills + Missing Skills

Match Percentage

The system calculates a skill match percentage based on the number of matched skills and the total number of required skills.

The general formula is:

Match Percentage = (Matched Skills / Required Skills) × 100

For example, if a career requires eight skills and three are matched, the calculated match percentage is based on three matched skills out of eight required skills.

Display of Results

The analysis results are displayed on the Skill Gap page. The user can view:

- Selected Career
- Current Skills
- Required Skills
- Matched Skills
- Missing Skills
- Match Percentage

This makes the result easier to understand and allows the user to identify the areas requiring additional development.

Database Integration

The analysis process can store the generated result in the MySQL database. This allows the latest analysis information to be retrieved later by the Career Report module.

Error Handling

If the user has not completed the required career selection or if analysis information is unavailable, the application can redirect the user to the appropriate previous stage and display a relevant message.

Importance of the Module

The Skill Gap Analysis Module forms the foundation for the personalized roadmap. The missing skills identified here are passed to the roadmap generation process.

Therefore, the overall relationship is:

Career Selection → Required Skills → Current Skills → Skill Comparison → Skill Gap → Personalized Roadmap

This module transforms general career requirements into user-specific skill information and provides the main analytical functionality of the application.

5.1.5 Skill Matching Module:

The Skill Matching Module is responsible for determining the degree of similarity between the user's current skills and the required skills associated with the selected career.

The module works closely with the Skill Gap Analysis Module. While the Skill Gap module identifies matched and missing skills, the Skill Matching Module provides a measurable representation of the user's current skill alignment.

Matching Process

The system receives the user's current skills and the required skills of the selected career.

The two sets are compared to determine which required skills are already present in the user's skill set.

For example:

Required Skills: HTML, CSS, JavaScript, Python
Current Skills: HTML, CSS

The matched skills are:

HTML, CSS

The missing skills are:

JavaScript, Python

Match Percentage Calculation

The matching percentage can be calculated using:

Match Percentage = (Number of Matched Skills / Total Number of Required Skills) × 100

This value provides a simple numerical representation of the current skill alignment.

Result Presentation

The result is displayed through the Skill Match interface. The system can present the matching information using percentages, cards, progress indicators, or other visual components.

The user can therefore understand their current skill position without manually calculating the result.

Relationship with Skill Gap

The Skill Matching Module and Skill Gap Analysis Module work together.

The matching process identifies what the user already possesses, while the gap analysis identifies what is still required.

Therefore:

Skill Matching → Matched Skills

Skill Gap Analysis → Missing Skills

Together, these results provide a complete view of the user's current career skill status.

Use in Roadmap Generation

The missing skills generated from the comparison are passed to the Personalized Roadmap Module. Therefore, the matching process directly influences the learning roadmap.

Use in Career Report

The match percentage and skill categories are also used by the Career Report Module to present a summarized result.

The Skill Matching Module therefore provides both analytical and practical value by converting the skill comparison into a measurable result that can be used throughout the application.

5.1.6 Personalized Roadmap Module:

The Personalized Roadmap Module converts the skill-gap analysis into a structured learning direction. It is one of the major features of the proposed system because it connects the identified missing skills with practical learning activities.

The roadmap is generated based on the user's selected career and current skill analysis rather than providing the same learning requirements to every user.

Roadmap Input

The primary input to this module is the list of **missing skills** identified during skill gap analysis.

The selected career role is also considered so that the roadmap remains connected to the user's career objective.

Personalized Generation

The system processes the missing skills and generates roadmap steps corresponding to the areas that require additional learning.

For example, if a user has already developed some of the required skills for Web Development but is missing other skills, the roadmap focuses on those missing areas.

This approach avoids unnecessarily recommending skills that the user has already identified as current skills.

Roadmap Structure

A roadmap step can contain:

- Skill or Topic Name
- Learning Description
- Learning Status
- Learning Resource Start
- Learning Option

The roadmap can present these steps in an organized sequence.

Learning Resources

Each roadmap skill can be associated with an external learning resource. The **Start Learning** button allows the user to open the corresponding learning material.

This provides a direct connection between:

Missing Skill → Learning Resource

Progress Representation

The roadmap interface can display learning progress using progress indicators and status information. Roadmap items may be represented using states such as not started, in progress, or completed depending on the implemented functionality.

Career Readiness

The roadmap is also connected with the user's skill analysis. The existing skill match percentage provides an indication of how many required skills are currently matched.

As the user develops additional skills, the analysis can be used as a reference for understanding changes in career readiness.

Integration

The Personalized Roadmap Module receives information from the Skill Gap Analysis Module.

The flow is:

Skill Gap Analysis → Missing Skills → Roadmap Generation → Learning Resources

This makes the roadmap personalized because it is based on the user's specific skill gap.

Advantages

The roadmap helps users:

- Identify what to learn.
- Focus on missing skills.
- Organize learning activities.
- Access relevant learning resources.
- Connect learning activities with a career goal.

The Personalized Roadmap Module therefore transforms analytical results into an actionable learning direction and forms an important part of the career development workflow.

5.1.7 Career Report Module:

The Career Report Module provides a consolidated summary of the user's career and skill analysis. It collects the relevant results generated by the previous modules and presents them in an organized format.

The report is designed to help users review their career selection and current skill status from a single page.

Report Information

The Career Report can contain:

1. User Information
2. Selected Career Role
3. Current Skills Required
4. Skills Matched Skills
5. Missing Skills
6. Skill Match Percentage

These details provide a summary of the user's current alignment with the selected career.

Latest Analysis

The system can retrieve the latest available skill analysis associated with the logged-in user. This ensures that the report reflects the most recent analysis information stored in the database.

Report Generation Process

The report generation process can be represented as:

User Session → Retrieve Latest Analysis → Process Analysis Data → Generate Report → Display Results

The Flask backend retrieves the relevant analysis information from MySQL and passes the data to the report template.

Skill Summary

The report presents the user's current skills alongside the required skills for the selected career. Matched and missing skills are displayed separately so that the user can understand the areas of alignment and improvement.

Match Percentage

The report also displays the calculated skill match percentage. This provides a numerical summary of the relationship between the user's current skills and the career requirements.

User-Friendly Presentation

The report is designed as a summary page rather than displaying raw database information. Cards, sections, progress indicators, and organized lists can be used to make the information easier to understand.

Integration with Other Modules

The Career Report Module depends on information generated by the Career Selection, Skill Gap Analysis, and Skill Matching modules.

The overall flow is:

Career Selection → Skill Analysis → Skill Matching → Analysis Storage → Career Report

The report therefore acts as the final summary of the user's career skill analysis.

The Career Report Module completes the application workflow by providing users with a consolidated representation of their selected career, current skill position, identified skill gaps, and matching results.

5.2 Form / User Interface Design:

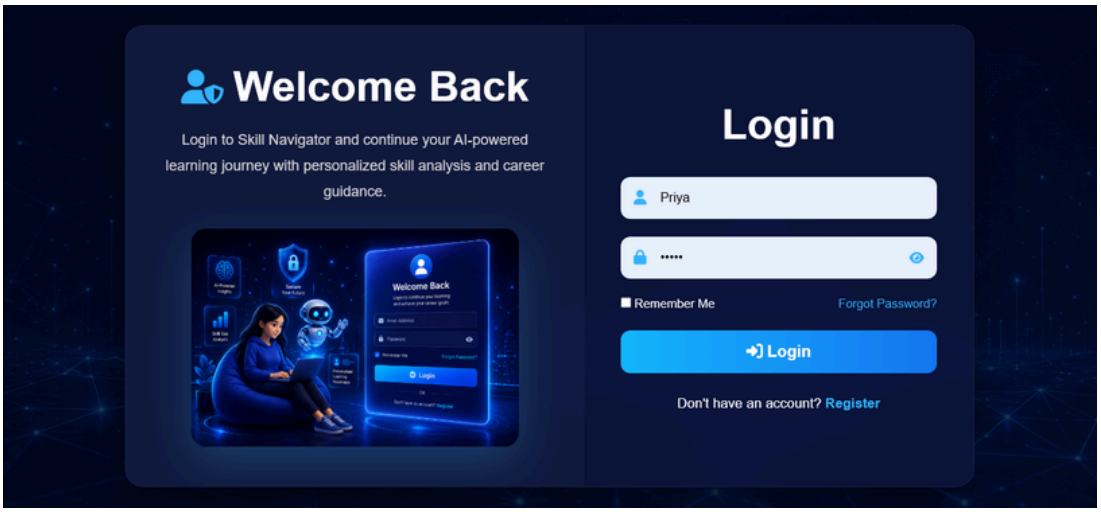
The Form and User Interface Design defines how users interact with the Skill Gap Analysis & Personalized Roadmap Generator. A simple and organized interface is important because the system contains several stages, including authentication, profile management, career selection, skill analysis, roadmap generation, and report viewing.

The frontend of the system is developed using **HTML and CSS**. Flask templates are used to display dynamic information received from the backend.

5.2.1 Login Interface

The Login page provides input fields for username and password. A login button allows the user to submit the credentials for authentication.

The interface provides appropriate feedback when login information is invalid.



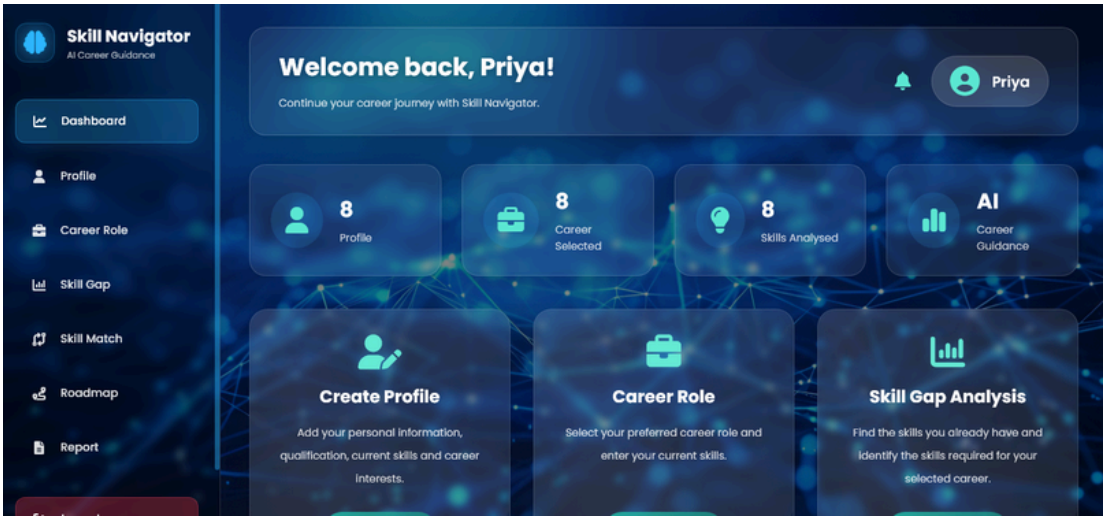
5.2.2 Dashboard Interface

The Dashboard acts as the central navigation point after successful login. It provides access to the major application sections.

The dashboard can display summary information and navigation options for:

1. Profile
2. Career
3. Skill Gap
4. Skill Match
5. Roadmap
6. Report
7. Logout

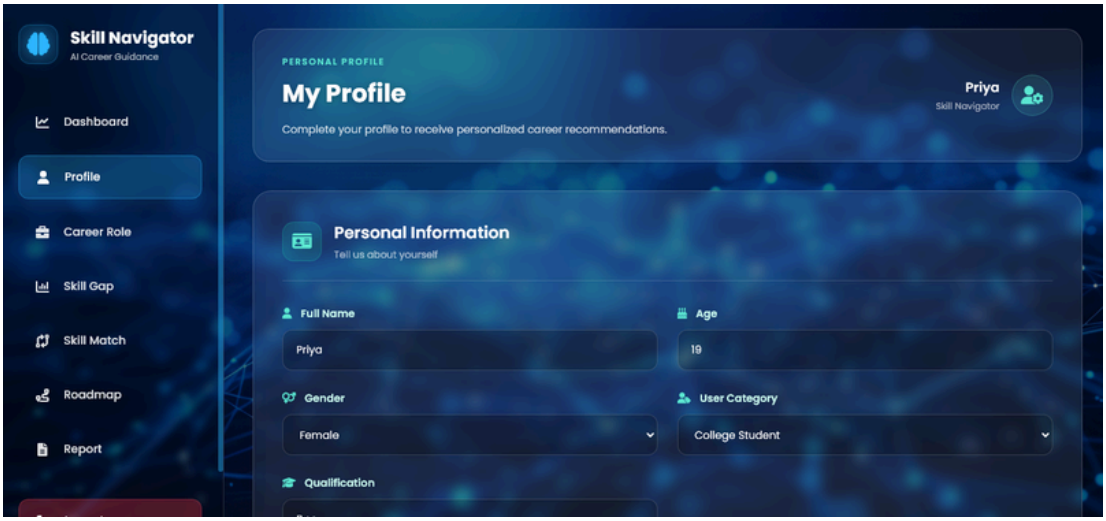
Dashboard Module



5.2.3 Profile Form

The Profile page contains form fields for entering and updating user information. The form is designed to make data entry simple and organized.

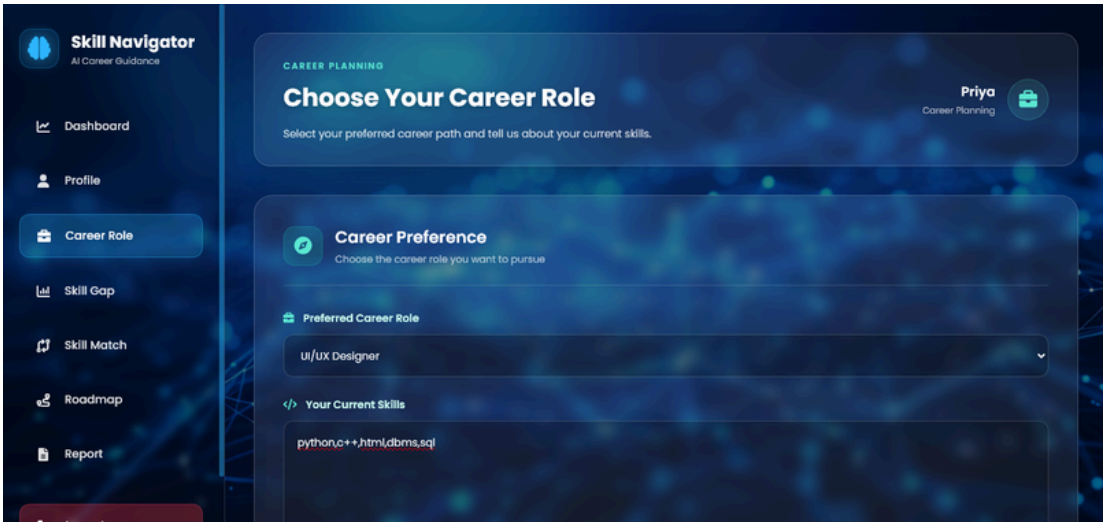
A Save button submits the information to the Flask backend.



5.2.4 Career Selection Interface

The Career page provides career selection options. The user selects the preferred career role and submits the selection.

After successful submission, the system provides confirmation and moves the user toward the Skill Gap Analysis stage.



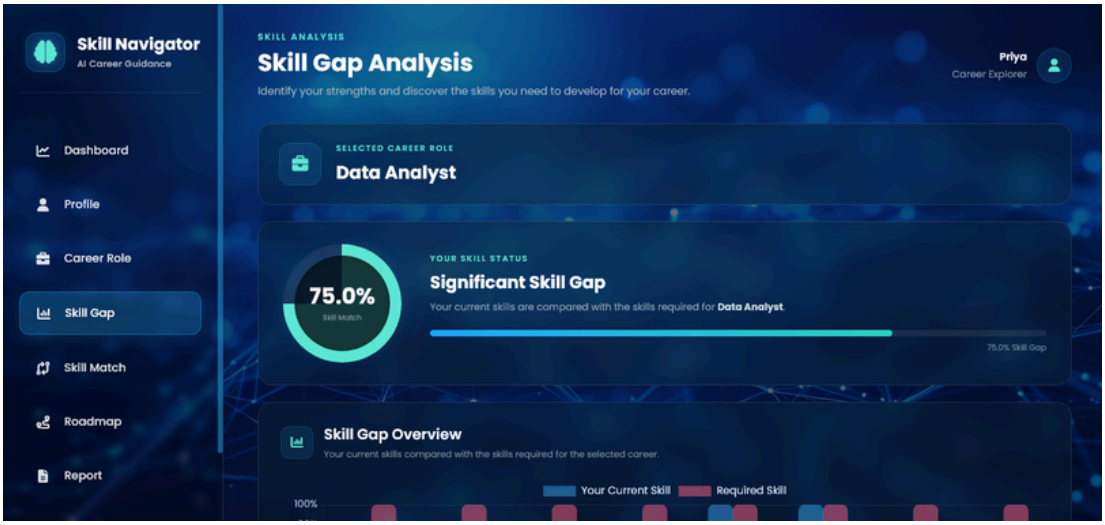
5.2.5 Skill Gap Interface

The Skill Gap page presents the results of the skill analysis.

Important information displayed includes:

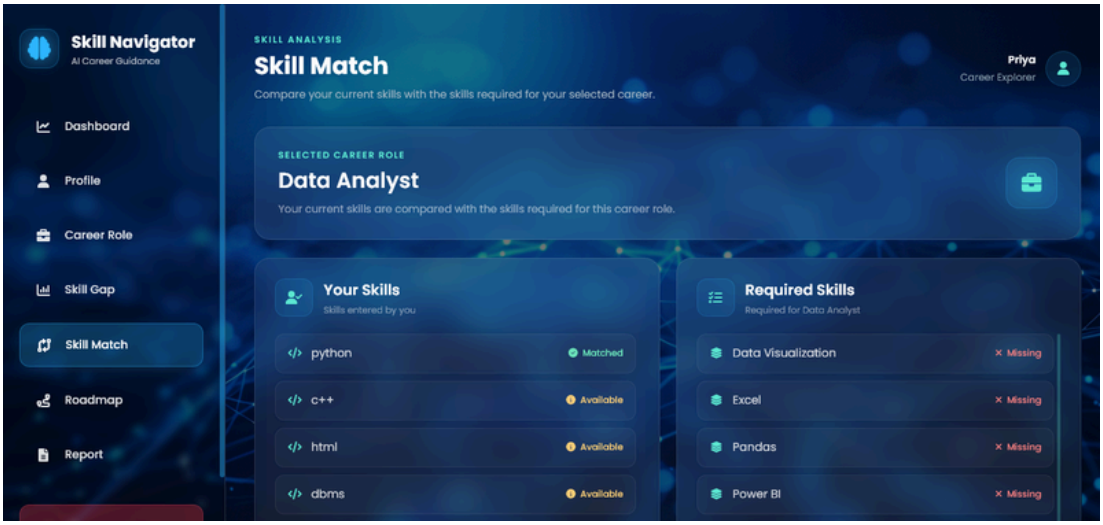
- Career Role
- Required Skills
- Current Skills
- Matched Skills
- Missing Skills
- Match Percentage

The information is organized using cards, lists, and visual indicators to make the analysis easier to understand.



5.2.6 Skill Match Interface

The Skill Match page focuses on the relationship between the user's current skills and career requirements. The match percentage can be displayed through a visual progress component along with supporting skill information.



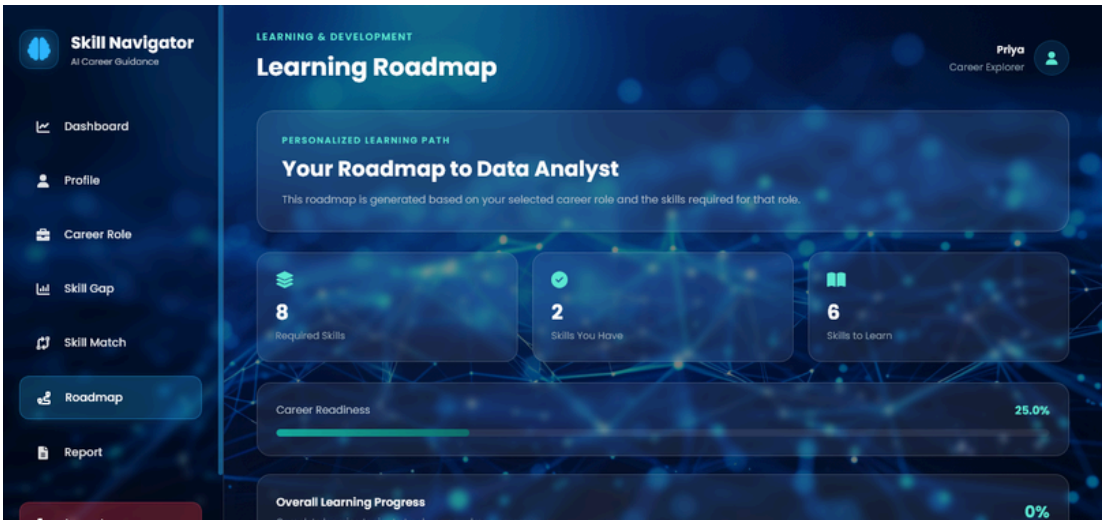
5.2.7 Roadmap Interface

The Roadmap page displays the personalized learning path. Each roadmap step represents a missing skill or learning area.

The interface can include:

- Roadmap title
- Career role
- Required skill summary
- Skills to learn
- Learning status
- Progress indicators
- Start Learning buttons

The Start Learning option provides access to an external resource related to the corresponding skill.



5.2.8 Career Report Interface

The Report page provides a summary of the complete analysis. The interface organizes the information into visually distinct sections.

The report can display the selected career, skill statistics, current skills, required skills, matched skills, missing skills, and match percentage.



5.2.9 Navigation Design

A consistent navigation structure is maintained across the application. A left-side navigation bar can provide access to the major pages.

This allows users to move between modules without having to return repeatedly to the starting page.

5.2.10 Visual Design

The application uses a modern interface design based on a dark background, translucent components, rounded cards, clear typography, and highlighted accent elements.

The glass-style interface helps separate important information while maintaining a consistent visual appearance across pages.

5.2.11 Form Validation and Feedback

Forms are designed to provide appropriate feedback after user actions. For example, after saving profile information or career preferences, the application can display a confirmation message.

Similarly, errors such as invalid login credentials or missing required information can be communicated to the user.

VI. SYSTEM TESTING

6.1 Testing Objectives:

System testing is an important stage in the development of the **Skill Gap Analysis & Personalized Roadmap Generator**. Testing is performed to verify that the developed application works according to the specified requirements and that the different modules function correctly both individually and as part of the complete system. The main purpose of testing is to identify errors, incorrect inputs, navigation problems, database issues, and unexpected system behavior before the application is used by end users.

The project consists of several interconnected modules such as User Login, Profile Management, Career Selection, Skill Gap Analysis, Skill Matching, Personalized Roadmap, and Career Report. Since the output of one module is used as input for another module, proper testing is necessary to ensure that information is transferred correctly throughout the application.

The first objective of testing is to verify the **functional correctness** of the application. Each feature must perform the operation for which it was designed. For example, the Login module must authenticate valid users and reject invalid login credentials. Similarly, the Profile Management module must store the user's profile information correctly, while the Career Selection module must save the selected career preference.

Another important objective is to verify the **accuracy of skill analysis**. The system compares the user's current skills with the skills required for the selected career. Testing ensures that matched skills and missing skills are identified correctly. The calculated skill matching percentage must also be generated based on the available skill information.

Testing also verifies the functionality of the **Personalized Roadmap Module**. The roadmap should be generated according to the user's identified skill gaps rather than displaying unrelated learning topics. The system should correctly identify the skills that need improvement and present appropriate learning steps.

The following are the major objectives of testing:

1. **To verify functional requirements:**

   Ensure that every implemented function performs according to the project requirements.

2. **To validate user authentication:**

   Verify that only valid users can access protected pages and that invalid login details are rejected.

3. **To verify data validation:**

   Ensure that required fields and user inputs are properly handled before processing.

4. **To verify database operations:**

   Confirm that user, profile, career preference, and analysis information can be stored and retrieved correctly.

5. **To validate skill gap calculation:**

   Ensure that the system correctly identifies matched and missing skills.

6. **To verify skill matching:**

   Check that the comparison between current skills and career-required skills produces meaningful results.

7. **To validate roadmap generation:**

   Ensure that the personalized roadmap is generated from the identified skill gaps.

8. **To verify navigation between modules:**

   Ensure that users can move correctly from Login to Dashboard, Profile, Career, Skill Gap, Skill Match, Roadmap, and Report.

9. **To verify session management:**

   Ensure that protected pages cannot be accessed without an authenticated user session.

10. **To validate report generation:**

    Verify that the Career Report displays the user's career, skill information, matching information, and roadmap-related results correctly.

11. **To identify and correct errors:**

    Detect programming, database, interface, and logical errors during development and testing.

12. **To verify usability:**

    Ensure that the application interface is understandable and that users can easily access the available functions.

Testing therefore provides confidence that the application performs its intended operations and that the complete workflow from user login to career analysis and personalized roadmap generation works in an organized manner.

6.2 Test Cases

Test cases are prepared to verify the behavior of different modules under different conditions. A test case specifies the module being tested, the input or condition provided, and the expected behavior of the system. Test cases help identify whether the application responds correctly to both valid and invalid inputs.

The following representative test cases cover the major functionalities of the **Skill Gap Analysis & Personalized Roadmap Generator**.

Test Case ID	Module	Test Condition / Input	Expected Result
TC-01	Login	Enter valid username and password	User should be authenticated and redirected to Dashboard
TC-02	Login	Enter incorrect username or password	Login should be rejected and an appropriate message should be displayed
TC-03	Login	Submit empty login fields	System should validate the input and prevent invalid submission
TC-04	Dashboard	Access dashboard after successful login	Dashboard should display the available navigation options
TC-05	Profile	Enter valid profile information and save	Profile information should be stored successfully
TC-06	Profile	Submit incomplete required information	System should validate the input
TC-07	Career Selection	Select a career role and save	Selected career preference should be stored
TC-08	Career Selection	Continue without selecting a career	System should prevent incomplete career selection
TC-09	Skill Gap Analysis	Analyze a user with a selected career and skills	Current, required, matched, and missing skills should be identified
TC-10	Skill Matching	Compare current skills with required skills	Matching information and percentage should be generated

TC-11	Skill Matching	User has several matching skills	Matching skills should be displayed correctly
TC-12	Skill Gap	User has missing required skills	Missing skills should be identified for further learning
TC-13	Roadmap	Generate roadmap after skill analysis	Personalized roadmap should be displayed based on identified skill gaps
TC-14	Roadmap	Open a learning resource	Relevant external learning resource should open
TC-15	Report	Access Report after analysis	Career and skill analysis information should be displayed
TC-16	Session Security	Access protected page without login	User should be redirected to the Login page
TC-17	Navigation	Move between project modules	Correct page should open for each navigation option
TC-18	Logout	Click Logout after login	User session should be terminated and login page should be displayed
TC-19	Database	Retrieve saved user information	Correct stored information should be retrieved
TC-20	Database	Save analysis information	Skill analysis information should be stored for later retrieval

Login Testing

Login testing verifies the authentication process. A valid username and password should allow the user to enter the application. Invalid credentials should not provide access to the protected pages. Empty fields and incorrect input formats should also be handled appropriately.

Profile Testing

Profile testing verifies whether user information is accepted and stored correctly. The system should ensure that the information entered by the user is transferred to the database and can be used by subsequent modules.

Career Selection Testing

The Career Selection module is tested to verify that the selected career role is correctly stored and used by the skill analysis process. The selected career should determine the required skills used for comparison.

Skill Gap and Matching Testing

Skill analysis is tested using different combinations of current and required skills. The system should correctly identify skills that are available in both sets and skills that are missing from the user's current skill set.

Roadmap Testing

The roadmap is tested after skill analysis has been completed. The system should generate learning steps based on the identified skill gaps. Learning resource buttons are also checked to ensure that the user can access the relevant external resource.

Report Testing

The Report module is tested to ensure that the latest available analysis information is displayed in an organized form. The report should represent the selected career and the associated skill analysis.

These test cases provide structured coverage of the major functions of the application. During actual execution, the test results can be recorded against each test case to document whether the expected behavior was obtained.

6.3 Unit Testing:

Unit testing is the process of testing individual components or modules of a software application separately. The main objective of unit testing is to verify that each individual component performs its intended operation before it is combined with other components.

In the **Skill Gap Analysis & Personalized Roadmap Generator**, the application is divided into several functional modules. Each module can be tested independently to identify errors at an early stage of development. Unit testing is particularly useful because an error in one module can affect the output of other modules when the modules are integrated.

User Login Unit Testing

The Login module is tested independently with valid and invalid credentials. When correct credentials are entered, the authentication process should successfully identify the user and create the required session information. When incorrect credentials are provided, access should be denied.

The login validation also checks whether empty fields are handled properly. This ensures that incomplete requests are not processed unnecessarily.

Profile Management Unit Testing

The Profile Management module is tested by providing valid profile information and checking whether it is correctly processed and stored. Individual profile fields are verified to ensure that the information received from the form is handled correctly.

The module is also tested with incomplete input to ensure that validation prevents incorrect or incomplete information from being saved.

Career Selection Unit Testing

The Career Selection module is tested independently to ensure that the selected career role is correctly received from the user interface and stored in the database. The selected career information is then checked to confirm that it can be retrieved for later analysis.

Skill Gap Analysis Unit Testing

The Skill Gap Analysis function is one of the major processing components of the project. It is tested independently using current skill information and career-required skill information.

The analysis process verifies:

- Current skills of the user.
- Required skills for the selected career.
- Skills available in both sets.
- Skills missing from the user's current skill set.
- Skill matching percentage.
- Career role associated with the analysis.

Testing this component independently helps ensure that the logical comparison is performed correctly.

Skill Matching Unit Testing

The Skill Matching component is tested using different combinations of current and required skills. For example, when some skills are common between the user's current skills and the required career skills, those skills should be identified as matched skills. Required skills that are not present in the user's current skill set should be identified as missing skills.

The matching calculation is also checked to ensure that the resulting percentage corresponds to the relationship between the user's available skills and the required skills.

Personalized Roadmap Unit Testing

The roadmap generation function is tested independently by supplying skill analysis information. The system should use the missing skills as the basis for generating learning steps.

The roadmap should not depend on unrelated default skills. Each learning step should correspond to a skill identified during the analysis process. Learning resource links are also checked to ensure that they are associated with the corresponding learning topic.

Career Report Unit Testing

The Career Report module is tested independently to ensure that the analysis information is correctly retrieved and displayed. The report should contain the relevant career role, skills, matching information, and other available analysis details.

Logout Unit Testing

The Logout function is tested separately to verify that the current user session is cleared and that the user is returned to the appropriate login page.

Benefits of Unit Testing

Unit testing provides several benefits to the project:

1. It identifies errors at an early stage.
2. It simplifies debugging.
3. It verifies the correctness of individual functions.
4. It reduces the possibility of errors during integration.
5. It improves the reliability of the application.
6. It makes maintenance easier.
7. It ensures that changes to one component can be tested independently.

Therefore, unit testing forms the foundation for reliable integration and system-level testing.

6.4 Integration Testing:

Integration testing is the process of combining individually tested modules and verifying whether they work correctly together. While unit testing checks each module independently, integration testing focuses on the communication and data flow between different modules of the application.

The **Skill Gap Analysis & Personalized Roadmap Generator** contains multiple interconnected modules. The output generated by one module becomes the input for another module. Therefore, integration testing is necessary to ensure that the complete workflow operates correctly without data loss, incorrect redirection, or inconsistent information.

The major integrations tested in the system are described below.

Login and Dashboard Integration

The first integration is between the User Login module and the Dashboard. After successful authentication, the system creates the user's session information and redirects the user to the Dashboard.

Testing verifies that:

Valid users can access the Dashboard.
User session information is maintained.
Invalid users cannot directly access protected pages.
Logout correctly terminates the active session.

This integration ensures that authentication and application access work together correctly.

Profile and Career Selection Integration

The Profile Management module is connected with the Career Selection module. After entering and saving profile information, the user can proceed to select a suitable career role.

Testing verifies that the profile information is saved successfully and that navigation from Profile to Career Selection occurs correctly. The selected career information must also be stored so that it can be used by subsequent analysis modules.

Career Selection and Skill Gap Analysis Integration

This is one of the most important integrations in the project. The selected career role determines the required skills that are used during skill gap analysis.

The integration process can be represented as:

Selected Career → Required Skills → Current Skills → Skill Comparison → Skill Gap

Testing verifies that the career selected by the user is correctly passed to the analysis process. The corresponding career-required skills should then be retrieved from the available career-skill dataset.

Skill Gap Analysis and Skill Matching Integration

The Skill Gap Analysis and Skill Matching modules work closely together. The analysis process identifies the user's current skills and compares them with the required skills for the selected career.

The integration verifies that:

Current skills are correctly retrieved.
Required career skills are correctly identified.
Common skills are classified as matched skills.
Unavailable required skills are classified as missing skills.
The matching percentage is generated from the comparison.

Correct integration between these components is important because the output of the skill comparison is used by the roadmap and report modules.

Skill Analysis and Personalized Roadmap Integration

The Personalized Roadmap module depends on the result of the skill analysis. The missing skills identified during analysis are used to generate learning steps.

The integration follows the sequence:

Skill Analysis → Missing Skills → Learning Topics → Personalized Roadmap

Testing verifies that the roadmap is generated according to the user's identified skill gaps. If a user has already acquired a particular required skill, that skill should not unnecessarily appear as a learning requirement.

Skill Analysis and Career Report Integration

The Career Report module uses the analysis information generated by the Skill Gap and Skill Matching processes. Testing ensures that the latest available analysis information is retrieved and represented correctly in the report.

The report should maintain consistency with the information shown on the Skill Gap and Skill Matching pages.

Roadmap and Learning Resource Integration

The roadmap contains learning steps associated with missing skills. Each learning step can provide access to an external learning resource relevant to that skill.

Testing verifies that the learning resource action corresponds to the appropriate learning topic and that the external resource can be opened through the provided link.

Navigation Integration

The complete navigation structure is also tested. The expected workflow is:

Login → Dashboard → Profile → Career Selection → Skill Gap → Skill Match → Roadmap → Report → Logout

Testing ensures that each navigation option opens the correct page and that the required user session and analysis information are maintained throughout the workflow.

Database Integration

Database integration testing verifies communication between the Flask application and MySQL database.

Operations such as inserting, updating, and retrieving user and analysis information are checked.

The system should retrieve the correct information for the currently logged-in user and should not mix information between different user accounts.

Importance of Integration Testing

Integration testing helps identify problems that may not appear during individual unit testing. It verifies that the modules communicate correctly and that data is transferred from one stage of the application to the next

For this project, integration testing is particularly important because the personalized roadmap and career report depend directly on the results generated by the skill analysis process. Successful integration therefore ensures that the application behaves as one complete career guidance system rather than as a collection of independent pages.

6.5 System Testing:

System testing is the process of testing the complete application as a single integrated system. After individual modules and their interactions have been tested, the entire **Skill Gap Analysis & Personalized Roadmap Generator** is evaluated to determine whether it satisfies the functional requirements of the project.

System testing considers the complete user journey, beginning with authentication and continuing through profile management, career selection, skill analysis, skill matching, personalized roadmap generation, career report generation, and logout.

Functional Testing

Functional testing verifies whether the application performs all the functions specified for the project.

The major functions tested include:

User registration/login-related access.
User authentication.
Dashboard access.
Profile information management.
Career selection.
Skill analysis.
Skill matching.
Missing skill identification.
Personalized roadmap generation.
Learning resource access.
Career report generation.
Logout and session termination.

Each function is tested with appropriate inputs to verify the expected output.

Input Validation Testing

Input validation is performed to ensure that incorrect or incomplete information does not cause unexpected system behavior.

Examples include:

Empty username or password fields.
Invalid login credentials.
Incomplete profile information.
Missing career selection.
Missing analysis information.
Accessing a module before completing the required previous step.

Validation ensures that the system responds appropriately rather than processing invalid data.

Session and Security Testing

Session testing verifies that the application correctly manages authenticated user access. Protected pages such as Profile, Career, Skill Gap, Skill Match, Roadmap, and Report should require a valid user session.

The testing process verifies that an unauthenticated user attempting to access a protected page is redirected to the Login page.

Logout functionality is also tested to ensure that the active session is terminated appropriately.

Database Testing

Database testing verifies that information is correctly stored and retrieved from MySQL.

The following database operations are considered:

1. User information storage.
2. Profile information storage.
3. Career preference storage.
4. Skill analysis storage.
5. Retrieval of the latest analysis information.
6. Updating relevant user information.
7. Maintaining user-specific records.

Database testing helps ensure that information displayed by the application corresponds to the appropriate stored data.

Skill Analysis Testing

Skill analysis is tested using different combinations of user skills and career-required skills.

For example, if the selected career requires several skills and the user possesses only a subset of them, the system should distinguish between the skills the user already has and those that need to be developed.

This testing verifies the logical accuracy of the skill comparison process.

Personalized Roadmap Testing

The Personalized Roadmap is tested using the output generated by the Skill Gap Analysis module. The roadmap should reflect the identified missing skills.

Testing verifies that:

Missing skills are considered for learning.
Existing skills are not unnecessarily treated as missing.
Learning topics correspond to the identified gaps.
Learning resources are associated with the appropriate topics.
Roadmap information is displayed clearly.

This ensures that the roadmap provides a logical continuation of the skill analysis process.

Report Testing

The Career Report is tested as the final information-summary component of the system. The report should display information obtained from the user's analysis.

The testing verifies consistency between:

Skill Gap Analysis → Skill Matching → Roadmap → Career Report

The report should represent the user's selected career and the corresponding skill analysis information without displaying unrelated user data.

User Interface Testing

The user interface is tested to verify that the application is visually organized and usable.

The following aspects are checked:

Sidebar navigation.
Page headings.
Buttons and links.
Forms.
Input fields.
Cards and information sections.
Skill information displays.
Progress indicators.
Roadmap sections.
Report sections.
Responsive layout behavior.

The interface should allow users to understand the available functions without unnecessary complexity.

Navigation Testing

Navigation testing verifies that users can move through the application without broken routes or incorrect redirects.

The primary navigation flow is:

Login → Dashboard → Profile → Career → Skill Gap → Skill Match → Roadmap → Report → Logout

Each route is checked to ensure that the correct page is displayed and that required session or analysis information is available.

External Learning Resource Testing

The learning resource links included in the roadmap are tested to verify that they correspond to the associated skill. When the user selects a learning option, the relevant external learning page should open.

This provides a connection between the application's skill-gap result and further learning activities.

Compatibility Testing

The web application can be tested using commonly available desktop web browsers to verify that the interface and functionality are displayed correctly. The project is primarily developed and tested in a Windows environment using a modern web browser.

Basic compatibility testing includes:

Page loading.
Navigation.
Forms.
Buttons.
Layout.
Text readability.
External links.

Overall System Validation

After completing the individual testing activities, the complete application workflow is evaluated from the user's perspective. A user should be able to log in, manage profile information, select a career, view skill analysis, identify skill gaps, access a personalized roadmap, view the career report, and finally log out.

The overall testing process confirms whether the different components work together as a complete system. Any errors identified during testing can be corrected before the final deployment or demonstration of the project.

Summary of System Testing

System testing provides final validation of the developed application. It ensures that the functional modules, database operations, navigation, skill analysis, roadmap generation, and reporting components operate together according to the intended design.

Through unit testing, integration testing, and system testing, the project can be evaluated at different levels. This structured testing approach improves the reliability, consistency, and usability of the **Skill Gap Analysis & Personalized Roadmap Generator**.

VII. CONCLUSION AND FUTURE ENHANCEMENT

7.1 Conclusion:

The **SKILL GAP ANALYSIS & PERSONALIZED ROADMAP GENERATOR** is a web-based career guidance application developed to help users understand the skills required for a selected career and identify the difference between their existing skills and the skills expected for that career. The system provides an organized approach to career-oriented skill analysis by bringing profile management, career selection, skill comparison, roadmap generation, and career reporting into a single application.

In the existing approach, users often depend on different websites, learning platforms, search engines, and manual comparisons to understand the skills required for a particular career. This process can require considerable effort and may make it difficult for beginners to determine which skills they should learn first. The proposed system addresses this problem by providing a structured workflow from career selection to skill-gap identification and personalized learning guidance.

The application begins with user authentication. After successful login, the user can manage their profile and select a preferred career role. The selected career is associated with the required skills available in the career- skill dataset. The system then considers the user's current skills and compares them with the required skills for the selected career.

The **Skill Gap Analysis Module** is an important component of the application. It identifies the skills that the user already possesses and the skills that are missing for the selected career. The system also calculates a skill matching percentage, providing the user with a simple numerical representation of the relationship between their current skills and the required skills.

The **Skill Matching Module** further organizes the comparison by presenting matched and missing skills. This makes it easier for users to understand their current position with respect to their selected career. Instead of requiring users to manually compare multiple skill lists, the application performs the comparison automatically.

One of the major features of the project is the **Personalized Roadmap Module**. The roadmap is generated using the skill-gap information obtained from the analysis. The identified missing skills are converted into learning steps so that users can focus on the areas where additional learning is required. Relevant external learning resources can also be accessed through the roadmap, allowing users to continue their learning outside the application.

The **Career Report Module** provides an organized summary of the user's career and skill analysis information. It allows the user to view the important results in a consolidated format. The report can help users understand their selected career, current skills, required skills, matched skills, and identified skill gaps.

The project uses **Python and Flask** for application development, **MySQL** for database management, and **HTML and CSS** for the user interface. A career-skill dataset is used to associate career roles with the skills required for those roles. The combination of these technologies provides a practical foundation for developing a database-driven career guidance web application.

The modular structure of the system also makes the application easier to understand and maintain. Each major function is implemented as a separate module, while the modules are connected through the application's workflow. Authentication controls access to the system, profile information provides user-specific details, career selection determines the analysis context, skill analysis identifies gaps, the roadmap provides learning direction, and the report summarizes the results.

Testing activities such as unit testing, integration testing, and system testing are important for validating the application. They help ensure that individual modules work correctly, that information is transferred correctly between modules, and that the complete workflow functions as expected.

Overall, the **SKILL GAP ANALYSIS & PERSONALIZED ROADMAP GENERATOR** demonstrates how web technologies and structured skill data can be used to create a career-oriented learning support system. The project provides users with a systematic method to identify skill gaps and organize their learning priorities according to a selected career role.

The project also provides a foundation for further development. Additional career roles, larger datasets, advanced recommendation techniques, progress tracking, and intelligent learning recommendations can be incorporated in future versions. Thus, the current system serves as a practical base for developing a more advanced and comprehensive career skill development platform.

7.2 Future Scope:

The current **SKILL GAP ANALYSIS & PERSONALIZED ROADMAP GENERATOR** provides the fundamental features required for career-oriented skill analysis and roadmap generation. However, the system can be enhanced with several additional features and technologies in the future. These improvements can make the application more adaptive, intelligent, and useful for a wider range of users.

1. AI-Based Career Recommendation

A future version of the system can include an AI-based career recommendation engine. Instead of requiring the user to select a career manually, the system could analyze the user's profile, education, interests, existing skills, and career preferences to suggest relevant career roles.

2. Expanded Career-Skill Dataset

The current system can be extended by adding more career roles and a larger collection of required skills. The dataset can include emerging technology careers, specialized technical roles, and other professional fields.

A larger dataset would allow users to analyze skill requirements for a wider range of career options.

3. Intelligent Skill Recommendations

Future versions can recommend specific skills based on the user's selected career and existing skill level. The system could prioritize missing skills according to their importance to the selected career.

This would allow users to understand not only **what** skills are missing but also **which skills should be learned first.**

4. Advanced Personalized Roadmap

The roadmap can be enhanced by generating a detailed learning sequence. Instead of displaying only missing skills, the system could arrange topics according to prerequisites and difficulty levels.

For example:

1. Basic Concepts
2. Intermediate Skills
3. Advanced Skills
4. Practical Project
5. Career Preparation

Such a structure could provide users with a more systematic learning path.

5. Learning Progress Tracking

A future version can include detailed progress tracking. Users could mark individual learning topics as started, in progress, or completed. The application could then calculate overall learning progress based on completed topics.

Progress information could be represented using progress bars, percentages, and learning statistics.

6. Job Description Analysis

The application could also support job-description analysis. A user could provide a job description, and the system could identify the technical skills and requirements mentioned in it.

The extracted requirements could then be compared with the user's current skills to generate a job-specific skill- gap analysis.

7. Certification Recommendations

The system could provide information about relevant certifications associated with selected career roles. Certifications could be organized according to skill areas and career requirements.

8.Real-Time Industry Requirements

The system can be integrated with current job market information to identify in-demand skills and provide users with updated industry requirements for their selected career roles.

BIBLIOGRAPHY:

1. Python Software Foundation, Python Documentation, Python Programming Language Documentation.

2. Pallets Projects, Flask Documentation, Documentation for the Flask Web Framework.

3. Oracle Corporation, MySQL Documentation, MySQL Reference Manual.

4. Mozilla Developer Network, HTML Documentation, Web development documentation and references.

5. Mozilla Developer Network, CSS Documentation, Cascading Style Sheets reference and web development guides.

6. Visual Studio Code Documentation, Visual Studio Code User Guide, documentation for source-code development and debugging.

7. Pandas Documentation, Pandas User Guide, documentation for data processing and analysis using Python.

8. General career and skill-development resources used for understanding career roles, technical skills, learning requirements, and skill-based career planning.

9. Career-skill dataset used in the project for mapping career roles with their corresponding required skills.

10. Standard software engineering concepts and references related to Software Development Life Cycle, System Analysis, System Design, Database Management, Software Testing, and Web Application Development.